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Stemming Methane: A Fast Lane for Climate Health

Abstract

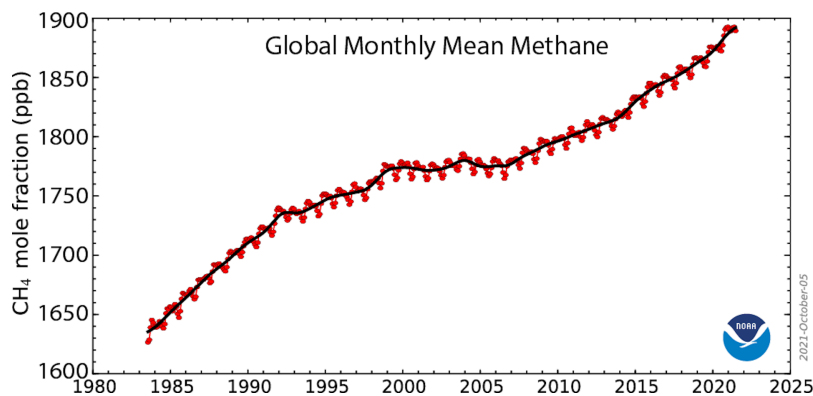
Carbon dioxide and methane are the two greenhouse gases that contribute most to global heating. Methane traps heat more powerfully than carbon dioxide but breaks down much faster, so cutting it can slow warming within years rather than decades. This lesson examines where methane comes from, how much of it could be cut by 2030, and how satellites are helping countries find leaks that go unreported.

Introduction

Climate change is mostly caused by human activities that change the makeup of the atmosphere and interfere with the natural flow of energy through the climate system. The results include heavier rains, stronger storms, more intense droughts, heatwaves, and wildfires. In 2020, methane in the atmosphere reached record levels, about 2.5 times higher than before the industrial era. At the COP26 climate summit, more than 100 nations representing 70% of the global economy joined the Global Methane Pledge, promising to cut methane emissions 30% by 2030. Climate scientist Kevin Trenberth explains why stopping methane leaks from oil fields, pipelines, and landfills is one of the quickest ways to buy time while the world works on the bigger, longer-term challenge of cutting carbon dioxide.

How Do Scientists Study This?

Scientists track methane in two complementary ways. First, monitoring stations measure methane concentration in the air, reported in parts per billion, and plot monthly values with a 12-month rolling average to reveal long-term trends. Second, countries report their estimated emissions by sector to the United Nations. When researchers compare the two, they find a gap: the amount rising in the atmosphere is larger than what countries report. New satellites help close that gap by detecting large leaks from space. Researchers such as Ilissa Ocko and colleagues then analyze each sector to estimate how much methane could be cut using technology that already exists, and whether those cuts would save or cost money.



FROM THE ARTICLE

Global monthly methane in parts per billion: monthly values (red) and 12-month rolling average (black). Adapted from NOAA, CC BY-ND.

Why cutting methane works fast

- 1 Leaks & vents**
Oil, gas, landfills and livestock release methane
- 2 Traps heat**
More potent than carbon dioxide
- 3 Short lifespan**
Breaks down far faster than carbon dioxide
- 4 Quick results**
Cutting leaks slows warming within years

Study the Process

Measure the air

Monitoring stations record methane levels every month. A 12-month rolling average smooths out seasons so scientists can see the long-term rise.

Compare with reports

Scientists check countries' reported emissions against what actually builds up in the atmosphere, revealing leaks that were never counted.

Locate the leaks

Satellites such as MethaneSat pinpoint emission sources from space, from oil and gas fields to farms, so they can be targeted and fixed.

What Did They Discover?

Oil and gas operations produce 26% of global methane emissions. Methane levels flattened between 2000 and 2008, but have risen again since, largely because of fracking and related oil and gas activity. Ocko's team found that 85% of oil and gas methane could be readily cut by 2030: 50% at a profit and another 35% with existing technology. For landfills, the figure is 80%. Livestock methane is harder to reduce: only 32% could be readily cut. Overall, using available technologies promptly could reduce expected global methane emissions by 57% in 2030. Satellite data also exposed dramatic events, such as 164 tonnes of methane escaping during a single hour of pipeline repairs in Russia.

Methane by the numbers

2.5×

pre-industrial methane level, a 2020 record

85%

of oil and gas methane readily cut by 2030

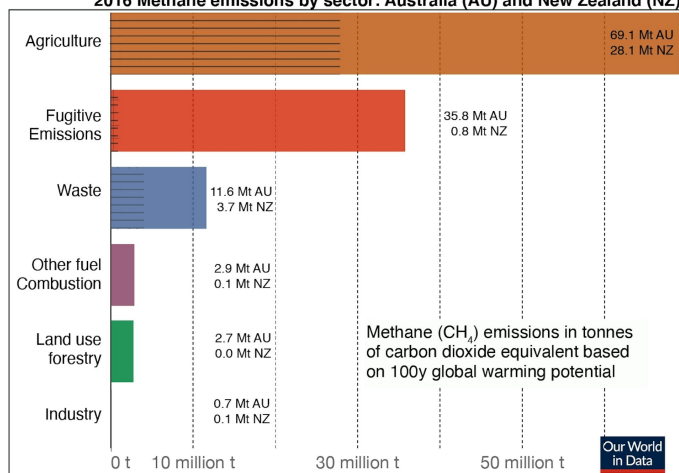
57%

possible cut in expected 2030 global emissions

The monitoring record shows methane was stable from 2000 to 2008, then began climbing again. That turning point lines up with the growth of fracking.

Look at the Evidence

2016 Methane emissions by sector: Australia (AU) and New Zealand (NZ)



FROM THE ARTICLE

2016 methane emissions by sector for Australia (AU) and New Zealand (NZ), in tonnes of carbon dioxide equivalent. Adapted from Our World in Data, CC BY-ND.

Takeaways

Fast lever

Methane is more potent than carbon dioxide but short-lived, so cutting it slows warming quickly.

Fixable leaks

Most oil, gas, and landfill methane can be cut by 2030, and many fixes pay for themselves.

Watch from space

Satellites reveal leaks countries fail to report, making emission pledges easier to check.

Key Terms

Methane

A powerful greenhouse gas and the main ingredient of natural gas. It traps more heat than carbon dioxide but breaks down in the atmosphere much faster.

Fugitive emissions

Gas that escapes unintentionally, for example from leaking pipelines, drilling operations, or old wells that were never properly sealed.

Biogenic methane

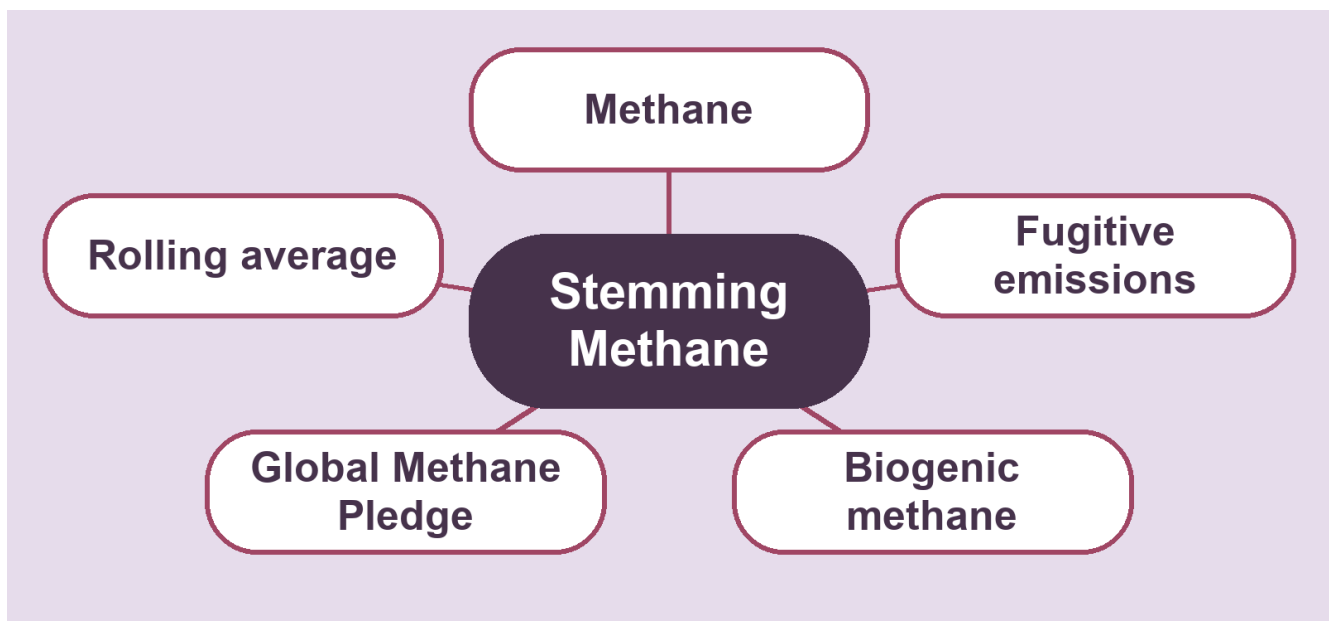
Methane produced by living things, such as the digestion of cattle and sheep. It is the main source of methane in New Zealand.

Global Methane Pledge

An agreement launched at COP26 in which more than 100 nations promised to cut methane emissions by 30% by 2030.

Rolling average

An average recalculated over a moving window, such as the last 12 months, used to smooth out short-term ups and downs in data.



Activity

Methane Budget Challenge

Students act as a climate task force deciding where to cut methane first, using the percentages from the article.

Materials: Sector cards: oil and gas, landfills, livestock; Calculator; Poster paper or a shared slide

1. In small groups, list the three methane sources described in the article and the share of each that could be readily cut by 2030.
2. Rank the sectors from easiest to hardest to reduce. Use the split between economically feasible and technically feasible cuts to justify your ranking.
3. Explain why methane cuts slow warming faster than equal efforts on carbon dioxide, using the words potent and lifespan.
4. Propose one way satellite monitoring could hold a country accountable to the Global Methane Pledge.
5. Present your plan and discuss: why does the article say cutting carbon dioxide is still the long-term challenge?

References

Trenberth, K. (2021). Stemming methane leaks from oil fields, pipelines and landfills could help us slow global warming quickly. *The Conversation*. <https://doi.org/10.64628/AA.dxm9gdpa>

Study by climate scientist Ilissa Ocko and colleagues (2021) on sector-by-sector methane cuts, as described in the article above.